Adaptive Cyber Deception, Human-AI Teaming, and Group Problem Solving

Yinuo Du, Ph.D

- School of Computer Science
- Carnegie Mellon University
- Supported by ARL-CRA and MURI-CATCH
- Human and AI Decision-Making
- in Cybersecurity:

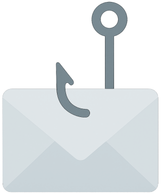
Yinuo Du, Ph.D (cont.)

AI INSTITUTE
FOR SOCIETAL
DECISION MAKING

The 2024 threat landscape is fast + adaptive

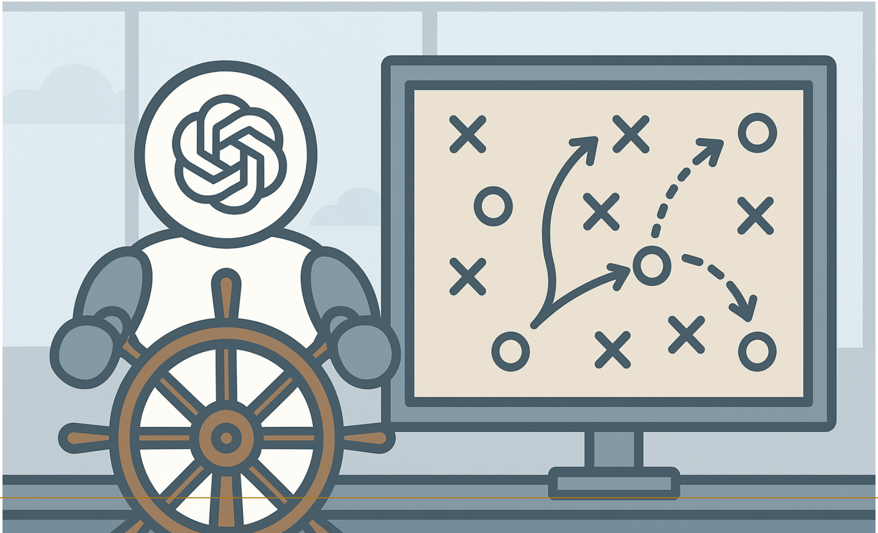
- Data Breach
- Phishing
- Ransomware
- 4.45 million
- 4.76 million

The 2024 threat landscape is fast + adaptive (cont.)

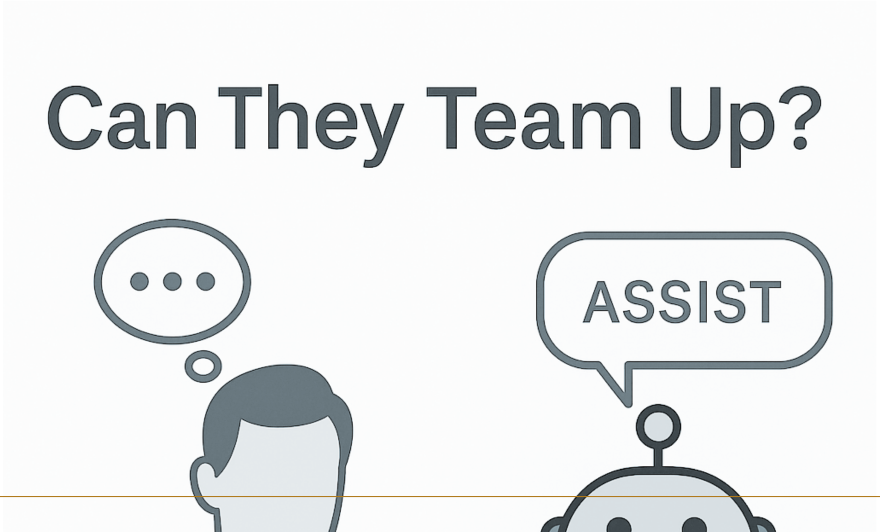


Too Fast for Humans

AI at the Helm?



Human-AI Cyber Defense: A Missing Partnership



Research

- Overview
- Game theory
- Reinforcement Learning
- Model of Human Cognition
- Human-AI Cyber Defense

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RL for Adaptive Cyber-Deception

- Human-AI Team Defense
- LLM for Group Problem-Solving
- Other projects
- Future vision
- Q&A

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Deception has been used to engage and mislead attackers

- Perturbation & Obfuscation
- Honey-X
- Pawlick, J., Colbert, E., & Zhu, Q. (2019). A game-theoretic taxonomy and survey of defensive deception for cybersecurity and privacy. ACM Computing Surveys (CSUR), 52(4), 1-28.
- True Configuration
- os]: Red Hat Linux 7

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Stackelberg security games have been used to model the strategic interaction between defender and attacker

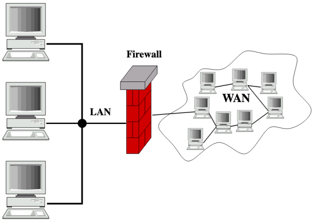
- Cyber Deception Game (Schlenker et al. 2018) models the defender's choice of deceptive network configurations and the attacker's reconnaissance as a zero-sum game.
- Cyber Camouflage Games (Thakoor et al. 2019) extends the model to general-sum to capture the uncertainty about attacker's payoff.
- Attack Graph Deception Game (Milani et al. 2020) models the sequential actions of the attacker as a path on the attack graph.

Stackelberg security games have been used to model the strategic interaction between defender and attacker

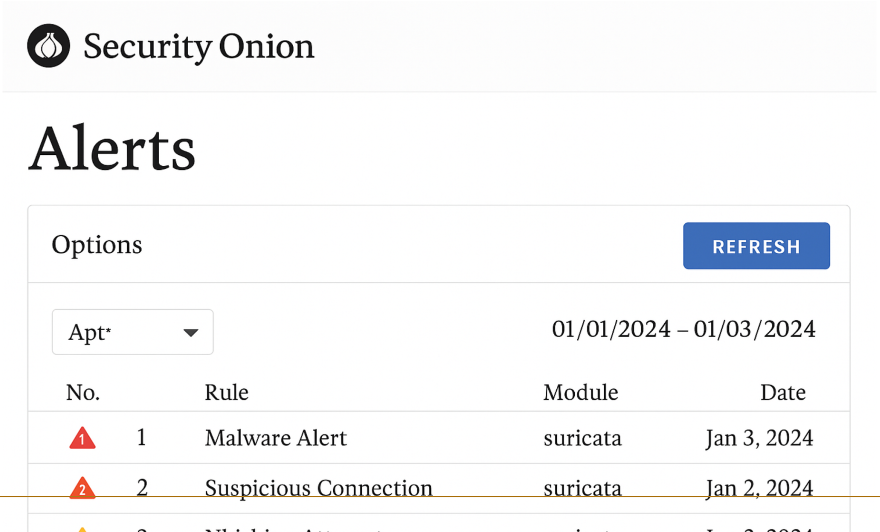
- T he defender and attacker's ability to adapt to real-time observations is under explored

Stackelberg security games have been used to model the strategic interaction between defender and attacker (cont.)

Attackers can gather more information about the network overtime



Defenders are also getting alerts in real time

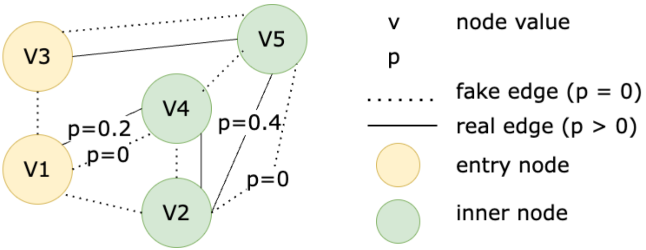


We propose an Adaptive Cyber Deception Game

- Observe
- Update defense strategy
- Surveillance
- Update attack strategy

We build our game model on an attack graph that can be interpreted at various granularity

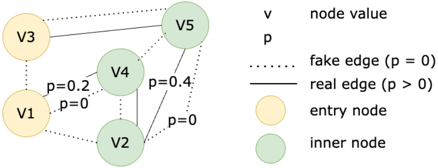
- probability of success



Defenders can take two types of action: deceptive and protective actions

- Deceptive
- Hide a real edge
- Add a fake edge
- Modify the perceived values of a set of nodes
- Protective

Defenders can take two types of action: deceptive and protective actions (cont.)



Each abstract action on the attack graph can be translated to multiple deceptive capabilities

- Hide real edge
- Add fake edge
- Change node value
- Fake foothold
- Fake privilege escalation

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We assume two types of adversary: Powerful and Naive

- Powerful
- Naive

We study the use of a reinforcement learning algorithm Proximal Policy Optimization (PPO) with self-play

- Input:
- Self location
- Perceived node values
- Real edge visibility
- Fake edge visibility

$$\pi_a(\boldsymbol{s}) = \boldsymbol{a}_a$$

We study the use of a reinforcement learning algorithm Proximal Policy Optimization (PPO) with self-play

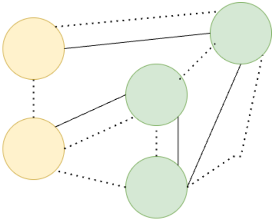
- Input:
- Attacker's location,
- Perceived node values,
- Real edge visibility,
- Fake edge visibility

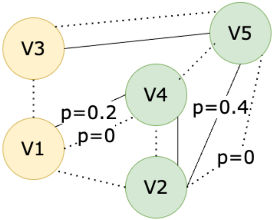
$$\pi_d(\boldsymbol{s}) = a_d$$

Reinforcement learning with self-play

- Attacker's
- Neural Network Policy
- Defender's
- 1. Collect experiences based on the player's current policies
- 2. Run one-step update for the defender/attacker's network parameters separately

Reinforcement learning with self-play (cont.)





$$\pi_a(s) = a_a$$

We compared the PPO defender with a heuristic defender

- Powerful/Naive)

Training Phase		Evaluation Phase	
Defender	Attacker	Attacker	Defender
PPO	PPO	PPO (PPO-trained)	PPO (trained) Heuristic
Heuristic	PPO	PPO (Heuristic-trained)	
		Heuristic	

We compared the PPO defender with a heuristic defender (cont.)

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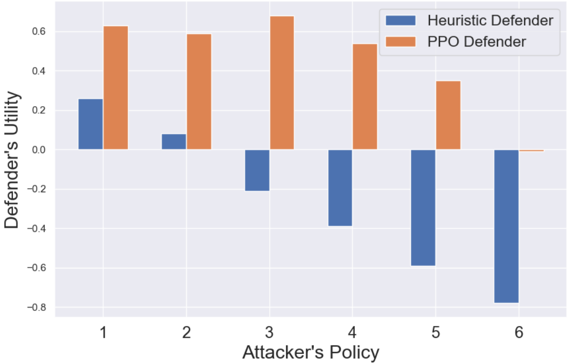
The PPO Defender always outperforms the heuristic defender.

- Naive
- Powerful

The PPO Defender always outperforms the heuristic defender.

(cont.)



We measure exploitability to see if the training reached nash equilibrium

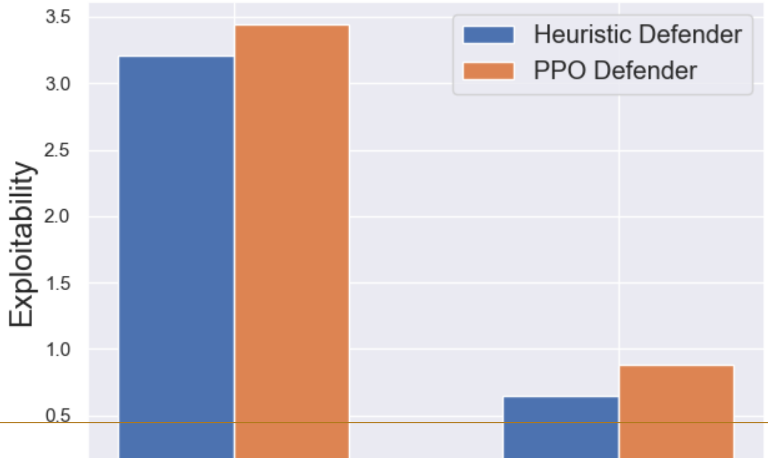
- Exploitability of
- Exploitability of attacker's policy
- Exploitability of defender's policy

$$Exp(\pi_a, \pi_d) = Exp_a(\pi_a, \pi_d) + Exp_d(\pi_a, \pi_d)$$

$$Exp_a(\pi_a, \pi_d) = U_a(\pi_a, \pi_d) - U_a(\pi_a, BR_d(\pi_a))$$

$$Exp_b(\pi_a, \pi_d) = U_d(\pi_a, \pi_d) - U_d(BR_a(\pi_d), \pi_d)$$

The (PPO defender, PPO attacker) pair of policies is closer to a Nash equilibrium



What now? With the promising performance in simulation, we went on deploying RL-based defense strategy in real-world network

rapid7/metasploit-framework

#18621 Add module f Backup Migration WI plugin (CVE-2023-...

2 comments

jheysel-r7 opened on December 15, 2023

Instances (11) Info

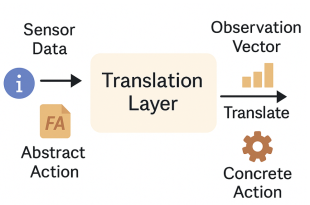
Find Instance by attribute or tag (case-sensitive)

Instance state = running | Clear filters

Name	Instance ID	Instance state
Public Server 1	i-0019dc99bc9bb85ea	Running
HoneyPot-Pub...	i-0f4012cd80d740b2c	Running
HoneyPot-NTP...	i-04a94d6d7fb9d3066	Running
PC1-WEB	i-07804fd9a42c65b49	Running
PC2-WEB	i-0013de65425006ac3	Running
PC3-WEB	i-0728c5e8f73936f4c	Running
Controller	i-04790eba4bba6c27a	Running
Public Server 2	i-031de534e971284af	Running
WEB	i-065a8db824659ae77	Running
NTP	i-0cf451c097f950db6	Running
DB	i-07ab0dcafa14c9bcc	Running

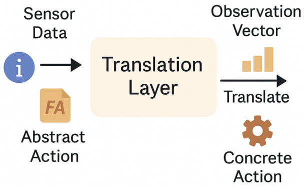
Bridging the Gap: Deploying RL Defenders on AWS

- Observation s



Bridging the Gap: Deploying RL Defenders on AWS

- Policy Improvement
- Reward
- Observation



Model-based reinforcement learning

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Latent Graph Dynamics and Opponent Belief Modeling for Strategic Planning

- Hafner, D., Lillicrap, T., Norouzi, M., & Ba, J. (2020). Mastering atari with discrete world models. arXiv preprint arXiv:2010.02193.

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- LLM for Group Problem-Solving
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Human-AI

- Team Defense
- Security Information and
- Event Management
- SIEM)
- Security Orchestration, Automation, and Response

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Existing AI techniques for cybersecurity are used as tools rather than teammates of humans

We propose a semi-supervisory Human-AI Teaming (HAT) paradigm

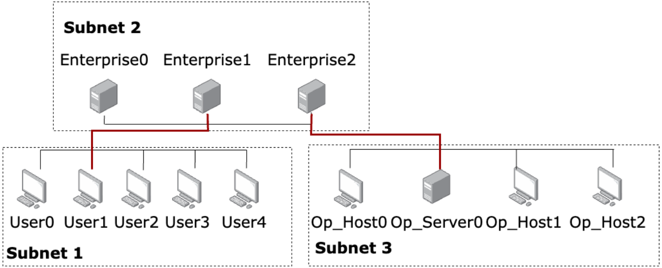
- Du, Y., Prbot, B., Malloy, T., Fang, F., & Gonzalez, C. (2025). Experimental evaluation of cognitive agents for collaboration in human-autonomy cyber defense teams. Computers in Human Behavior: Artificial Humans, 100148.

REDRAW: this slide's diagram was built from 1 PowerPoint shapes, which carry no image to extract.

The HAT defend an enterprise network together

- Du, Y., Prbot, B., Malloy, T., Fang, F., & Gonzalez, C. (2025). Experimental evaluation of cognitive agents for collaboration in human-autonomy cyber defense teams. Computers in Human Behavior: Artificial Humans, 100148.

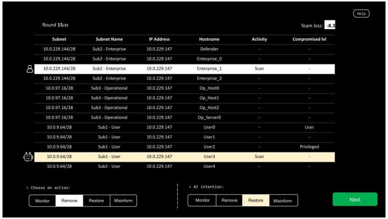
The HAT defend an enterprise network together (cont.)



HAT's goal: minimize cumulative loss

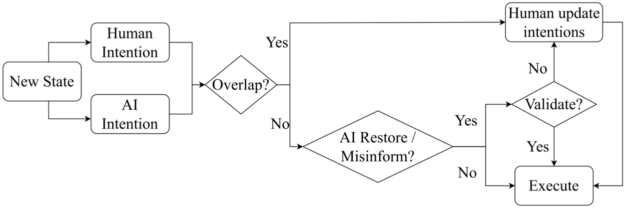
Event or Action	Defender's Loss
Attacker has administrator access on a Host	0.1
Attacker has administrator access on a Server	1
Attacker successfully Impact Op_Server0	10
Defender restore a Host or Server	1
Defender deploy a honeyfile on a host	0.5

Slide 36


Help
Round 15/25
Team loss: -8.3
Subnet | Subnet Name | IP Address | Hostname | Activity | Compromised lvl
10.0.229.144/28 | Sub2 - Enterprise | 10.0.229.147 | Defender | - | -
10.0.229.144/28 | Sub2 - Enterprise | 10.0.229.147 | Enterprise_0 | - | -
10.0.229.144/28 | Sub2 - Enterprise | 10.0.229.147 | Enterprise_1 | Scan | -
10.0.229.144/28 | Sub2 - Enterprise | 10.0.229.147 | Enterprise_2 | - | -
10.0.97.16/28 | Sub3 - Operational | 10.0.229.147 | Op_Host0 | - | -
10.0.97.16/28 | Sub3 - Operational | 10.0.229.147 | Op_Host1 | - | -
10.0.97.16/28 | Sub3 - Operational | 10.0.229.147 | Op_Host2 | - | -
10.0.97.16/28 | Sub3 - Operational | 10.0.229.147 | Op_Server0 | - | -
10.0.9.64/28 | Sub1 - User | 10.0.229.147 | User0 | - | User
10.0.9.64/28 | Sub1 - User | 10.0.229.147 | User1 | - | -
10.0.9.64/28 | Sub1 - User | 10.0.229.147 | User2 | - | Privileged
10.0.9.64/28 | Sub1 - User | 10.0.229.147 | User3 | Scan | -
10.0.9.64/28 | Sub1 - User | 10.0.229.147 | User4 | - | -
> Choose an action:
Monitor
Remove
Restore
Misinform
> AI intention:
Monitor
Remove
Restore
Misinform
Next

Human is responsible for resolving conflicts and validating risky AI actions

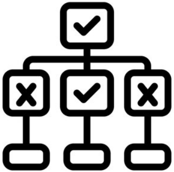
- Du, Y., Prbot, B., Malloy, T., Fang, F., & Gonzalez, C. (2025). Experimental evaluation of cognitive agents for collaboration in human-autonomy cyber defense teams. Computers in Human Behavior: Artificial Humans, 100148.



We compared HATs with cognitive agent, heuristic agent, and random agent

- Du, Y., Prbot, B., Malloy, T., Fang, F., & Gonzalez, C. (2025). Experimental evaluation of cognitive agents for collaboration in human-autonomy cyber defense teams. Computers in Human Behavior: Artificial Humans, 100148.
- Cognitive
- Heuristic
- Random

We compared HATs with cognitive agent, heuristic agent, and random agent (cont.)



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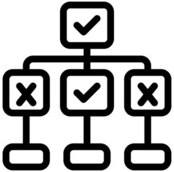
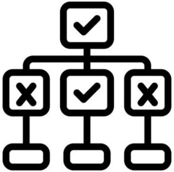
We compared HATs with cognitive agent, heuristic agent, and random agent (cont.)



We compared HATs with cognitive agent, heuristic agent, and random agent

- Cognitive
- Heuristic
- Random
- Du, Y., Prbot, B., Malloy, T., Fang, F., & Gonzalez, C. (2025). Experimental evaluation of cognitive agents for collaboration in human-autonomy cyber defense teams. Computers in Human Behavior: Artificial Humans, 100148.
- Adaptive, Competent

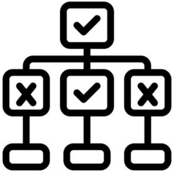
We compared HATs with cognitive agent, heuristic agent, and random agent (cont.)



We recruited human participants from Amazon Mechanical Turk (MTurk) to team up with the agents

- Cognitive 48 Participants
- Heuristic 42 Participants
- Random 66 Participants
- Du, Y., Prbot, B., Malloy, T., Fang, F., & Gonzalez, C. (2025). Experimental evaluation of cognitive agents for collaboration in human-autonomy cyber defense teams. Computers in Human Behavior: Artificial Humans, 100148.

We recruited human participants from Amazon Mechanical Turk (MTurk) to team up with the agents (cont.)



Experiment Procedure

- Consent form, Demographics
- Instructions,
- Quiz, Practice
- Cognitive Condition
- Heuristic Condition

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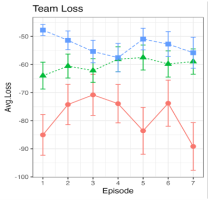
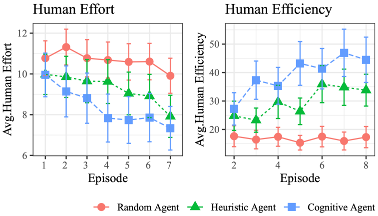
Cognitive partner had the highest performance than alternatives

- Cognitive Agent
- Team loss
- Avg. loss
- Episode



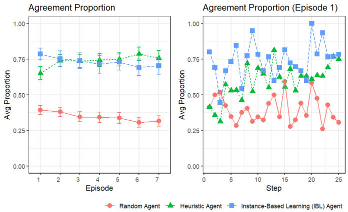
Team Loss Comparison

- IBL teammate agent had the highest performance across alternatives.
- IBL team loss somewhat decreased over time, heuristic somewhat increased.



Humans quickly learn to trust competent agents

- Cognitive Agent
- Agreement Proportion
- Avg. Proportion
- Episode



Humans working with cognitive agents achieve more while doing less

Perception of agents is affected by high expectations, which later led to blind trust or great disappointment

- It was challenging because you had to work with an AI that can not communicate. Made it difficult to come up with a plan or strategize. ”
- I trusted the partner more than I trusted myself. I was glad it was AI, I assumed they were using an algorithm to make the decisions.”

What now? Cyber defense requires coordinated, multi-step strategies

- Strategic collaboration is crucial
- A deeper understanding of task dynamics is needed

Teamwork

- Coordination
- Communication
- Disruption & Recovery
- In our ongoing work, we expand TDG platform to enable automated generation of team tasks and supports experiments with real agents.
- Team Task Generator

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RL for Adaptive Cyber-Deception

- Human-AI Team Defense
- LLM for Cyber Group Consensus
- Other projects
- Future vision
- Q&A

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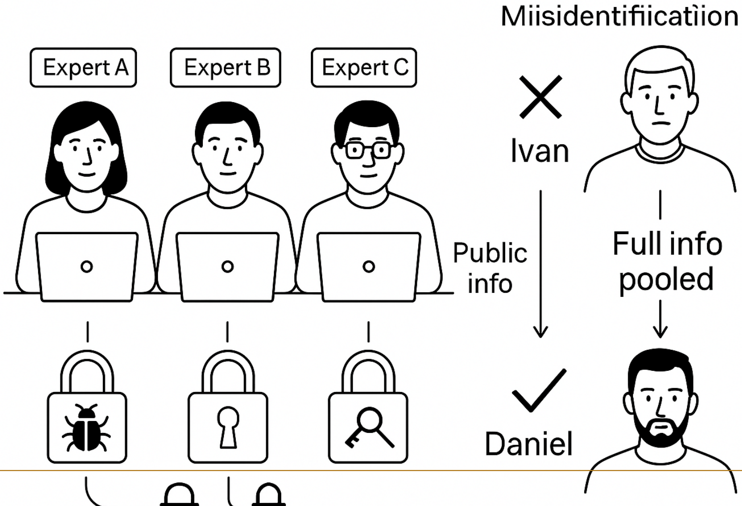
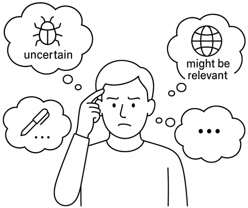
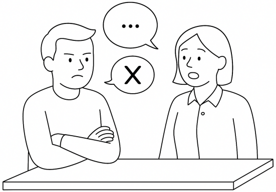
Attacker profiling is a group task



Individuals often fail to recognize the relevance of their own knowledge and are reluctant to express disagreement





We start by exploring similar group problem solving tasks with public datasets – Winter Survival Task

A peek into the human group conversation

- Braley, M., & Murray, G. (2018). The group affect and performance (GAP) corpus. In Proceedings of the group interaction frontiers in technology (pp. 1-9).

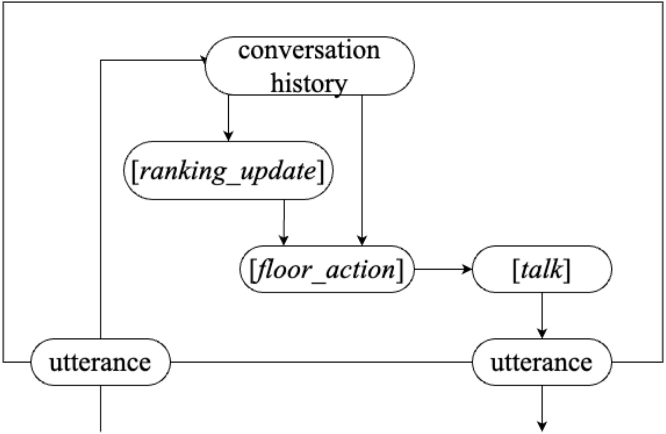
A peek into the human group conversation (cont.)

So what did everyone do as one"	Pink
I did uh cigarette lighter"	Blue
For one"	Blue
Mm okay I did knife"	Pink
Knife"	Green
Okay well do knife then"	Blue
Yeah yeah makes sense"	Blue
Well I think it that and and um the cigarette lighter are all pretty important"	Pink
So I would say cigarette lighter is two"	Pink
I didn’t write that down but I kinda made a connection afterwards"	Pink
Yeah okay"	Blue

we utilized the 0613 version of GPT-4-32k

- speaker id, text)
- Update ranking based on new information
- Decide whether to grab the conversation floor
- Generate natural language to express its opinion
- Can LLM agents do better? We designed a large language model based agent for this task

we utilized the 0613 version of GPT-4-32k (cont.)



We proposed an algorithm to enable free-form conversation in groups of LLM agents

- Start
- Wait for an agent to Grab the floor and become the speaker
- Is the conversation floor free?
- Is the discussion finished?
- The most eager preemptor become the next speaker
- Is anyone Claiming the floor ?

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Capturing How Humans Communicate and Feel

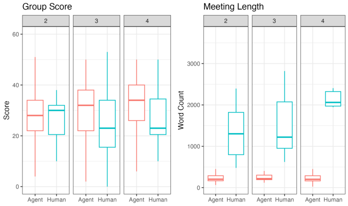
- Dialogue Acts
- Proposal
- Agreement
- Disagreement
- Confirmation

Metrics

- Performance Metrics
- Group Score — distance from expert ranking
- Meeting Length — total word count or time
- Affective Metrics
- Positivity / Negativity — based on sentiment annotations

Agent groups achieved higher scores in shorter time frames

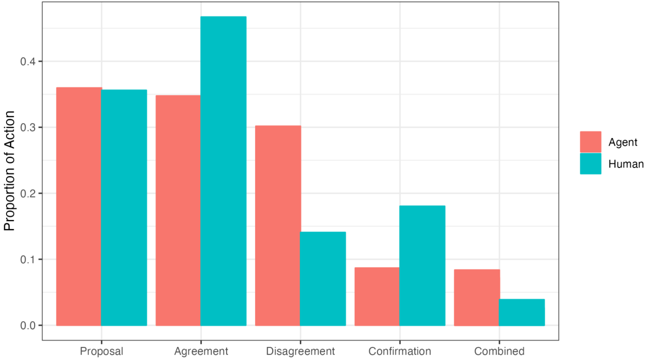
- Group Score
- Meeting Length
- Score
- Word COunt



Agent discussions reveal greater disagreement among agents within a group compared to human groups.

- Agent
- Human
- Proposal
- Agreement
- Disa greement

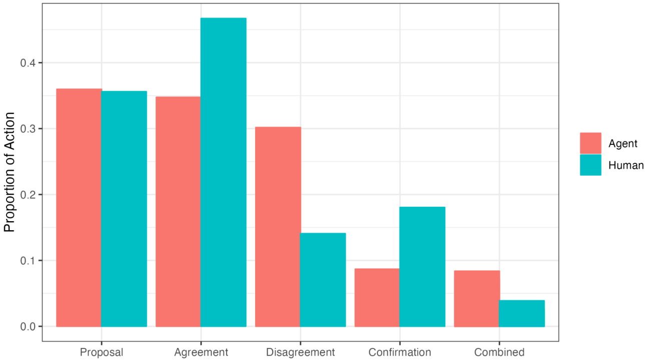
Agent discussions reveal greater disagreement among agents within a group compared to human groups. (cont.)



Agent groups also tend to produce more complex statements that contain both agreements and disagreements.

- Agent
- Human
- Proposal
- Agreement
- Disagreement

Agent groups also tend to produce more complex statements that contain both agreements and disagreements. (cont.)

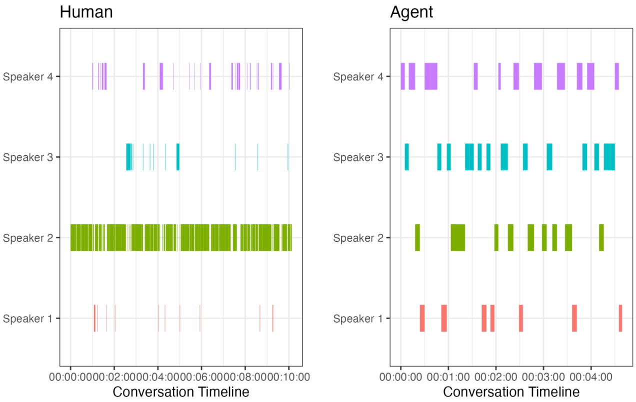


Agents participated in the discussion more equally than humans

- Speaker 1
- Speaker 2
- Speaker 3
- Speaker 4
- Agent

Agents participated in the discussion more equally than humans

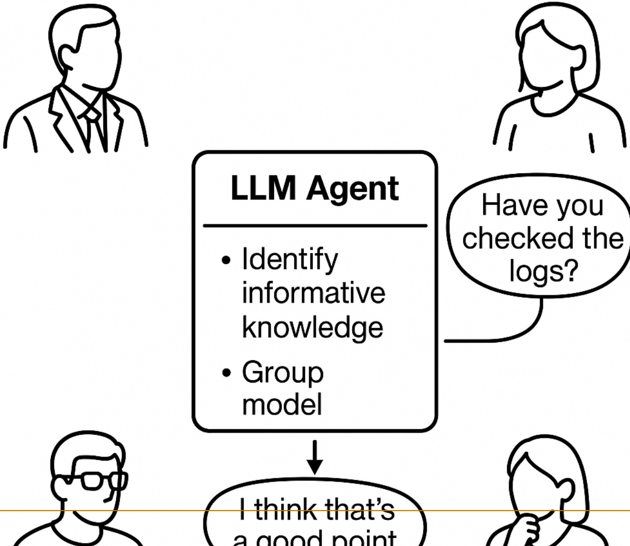
(cont.)



Moving Forward: LLM Agent -assisted group problem solving – Detecting Knowledge Gap & Group Dynamics

- Identify knowledge gap
- Model the group dynamic

Moving Forward: LLM Agent -assisted group problem solving – Detecting Knowledge Gap & Group Dynamics (cont.)



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- Human-AI Team Defense
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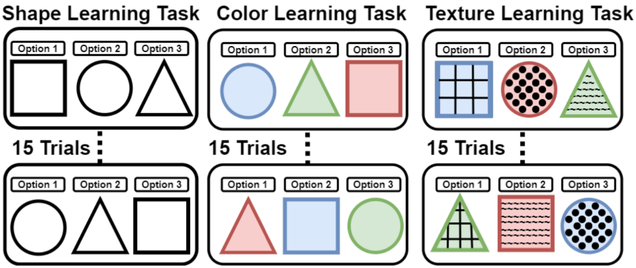
Contextual Bandits & Human Transfer-Learning Modeling

- Malloy, T., Du, Y., Fang, F., & Gonzalez, C. (2023, November). Accounting for Transfer of Learning Using Human Behavior Models. In Proceedings of the AAAI Conference on Human Computation and Crowdsourcing (Vol. 11, pp. 115-126).

Contextual Bandits & Human Transfer-Learning Modeling (cont.)

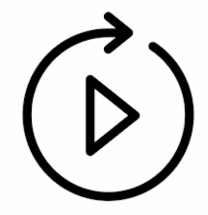


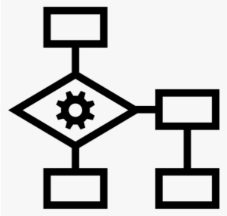
Stackelberg Security Games & Human Adversary Modeling

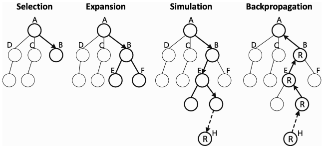
- Du, Y., Prebot, B., Malloy, T., & Gonzalez, C. (2024). A Cyber-War Between Bots: Cognitive Attackers are More Challenging for Defenders than Strategic Attackers. ACM Transactions of Societal Computing.

	Attack Replay	Rule-based Playbook	Monte-Carlo Planning	Ethical
Customizable				
Adaptive				
Scalable				

Stackelberg Security Games & Human Adversary Modeling (cont.)



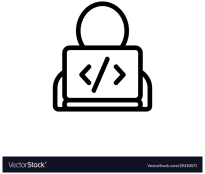


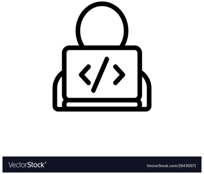


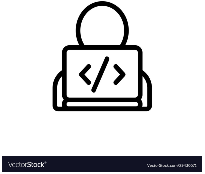
Iterated Prisoner's Dilemma & Human Groupmate Modeling

- Share?
- Not Share?
- Du, Y. , Singh, K., Aggarwal, P., Fang, F., & Gonzalez, C. Emergent Cooperative Decision-Making in Triadic Prisoner's Dilemma: Effects of Incentives and Information. Available at SSRN 5136406.

Iterated Prisoner's Dilemma & Human Groupmate Modeling (cont.)







RL for Adaptive Cyber-Deception

- LLM for Group Decision-Making
- Human-AI Team Defense
- Other projects
- Future vision
- Q&A

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Research Vision

- Research
- Vision
- Game theory
- Deployment Efficient Reinforcement Learning
- Computation Rational Model of Human Cognition
- Human-AI Cyber Defense

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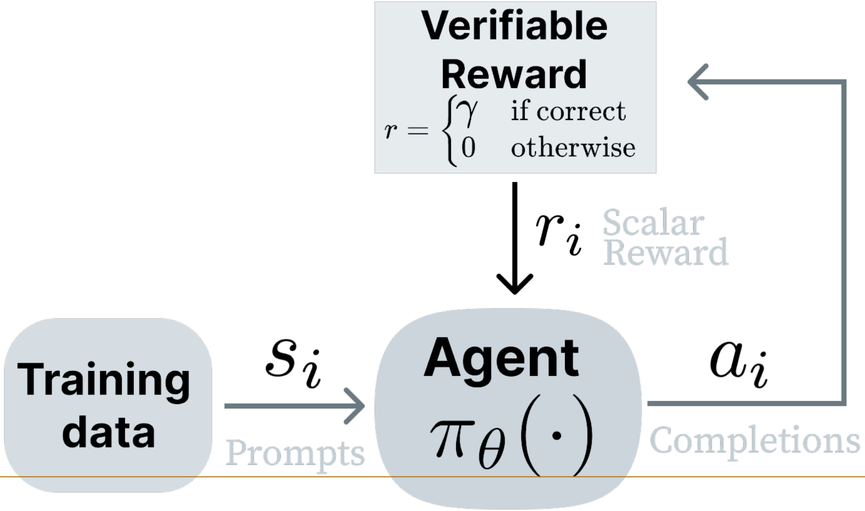
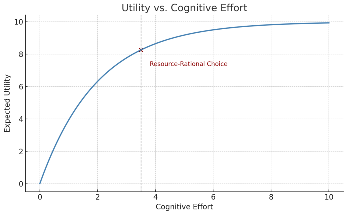
Deployment-Efficient Reinforcement Learning for Autonomous Defense



Computation Rational Behavior Models for Behavioral Cybersecurity

- Cognitive Effort
- Expected Utility



$$\max_{\pi} \mathbb{E}\left[\text{Utility}(\pi) - \lambda \cdot \text{Cost}(\pi)\right]$$

Thank you! Questions?

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Funding opportunities

- Grant experience
- PhD research supported by MURI-CATCH, and ARL-CRA
- Participated in proposal writing for
- IARPA-ReSCIND
- Cylab Future Enterprise Seed Grant

Backup Slides

What now ? Cognitive attacker can capture human learning behavior, but not the biases

- Default Setting Bias
- Sunk Cost Fallacy

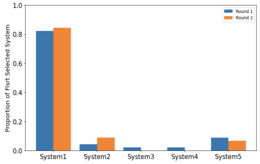

Figure 1. Proportion of participant's first system selection during the network reconnaissance phase.

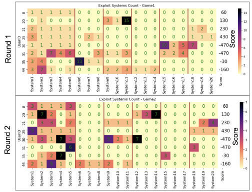

Figure 2. The number of attempts to exploit systems in Round 1 and Round 2.

In our ongoing work, we build computationally rational models of human behavior constrained by their internal environment, that is, cognition.

- Policy π
- Cognitive Resources
- (θ)
- Controller
- Internal

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Composition Reasoning, Causal Inference, & Optimal Forgetting

$$R(x) = \frac{1}{n} \sum_{i=1}^{n} R_i(x[S_i])$$

$$Q(x[S_i]) \leftarrow Q(x[S_i]) + \alpha(Q(x[S_i]) - \Phi_i o_{m,n})$$

$$\Phi_i \leftarrow \bar{\Phi}_i + \omega(\bar{\Phi}_i - o_{m,n})$$

Dynamic Weighting, Category Learning, & Contrast Effect